

AI3613 Database System Concepts Homework 3

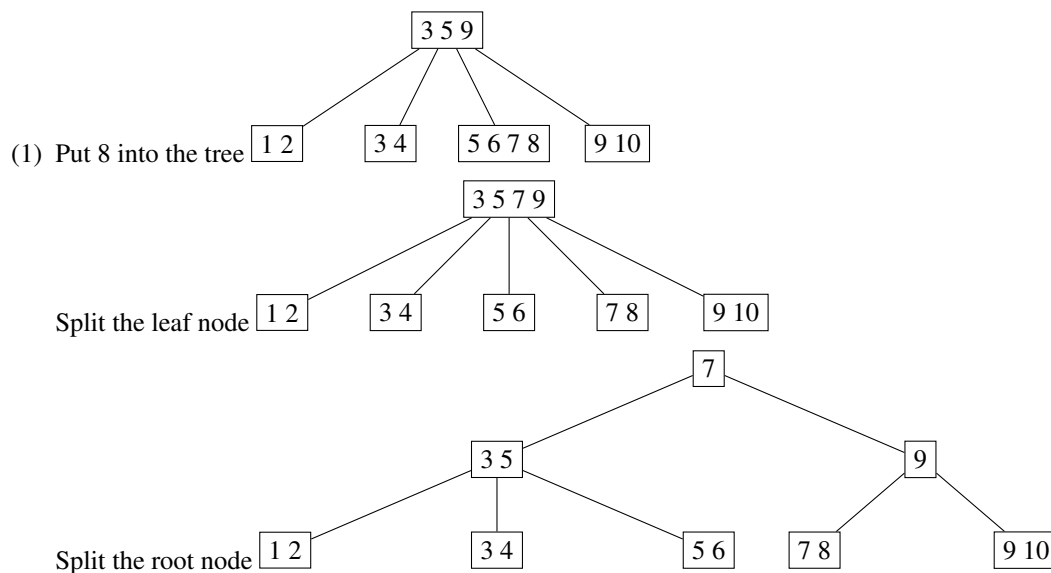
Kailing Wang 521030910356

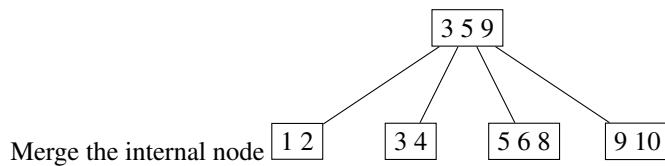
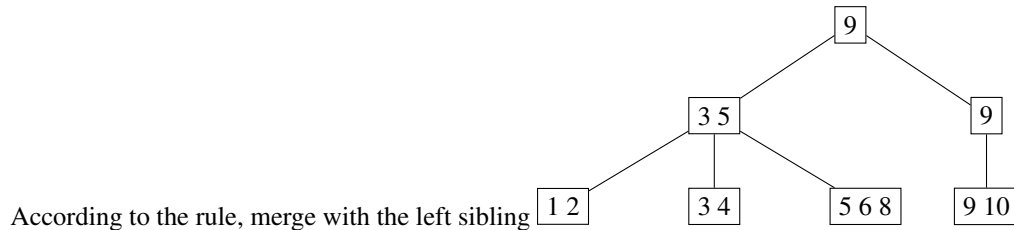
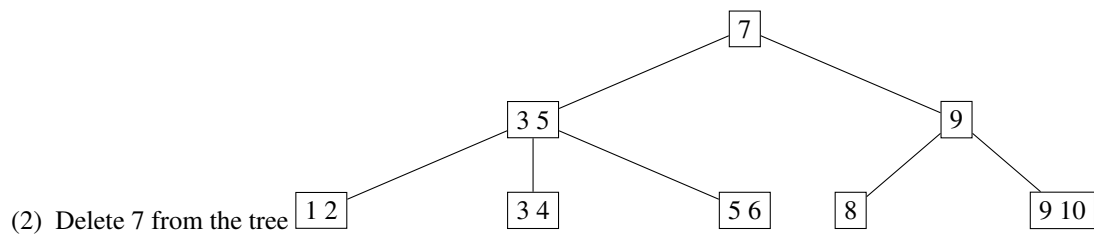
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Problem 1

- In this case, all 300 pages are needed to read from disk. Every page can contain tuples where there are students with gpa 3.7.
 - 100 pages are needed to read from disk. We only have to go over the gpa row and count. We do not need to know the name and id.
- In this case, we'd stop on finding a tuple with id=123. The most lucky case is that the first page contains the tuple. The worst case is that the last page contains the tuple. The pages we need to read from disk range from 1 to 300.
 - Similarly, we first scan the id row to find the offset for id=123. The pages we need to read from disk range from 1 to 100. Then we read two more pages to get the name and gpa. The total pages we need to read from disk range from 3 to 102.

Problem 2





- (3) 1.read root node, 1 page
 2. find the leaf node that contains the key 2, 1 page
 3. use the tail pointer to find the (3, 4) node, 1 page
 4. again to the 5, 6, 8 node, 1 page
 In total, 4 pages are needed to read from disk.

Problem 3

How long does it take you to finish the assignment? Give a score (1,2,3,4,5) to the difficulty of each problem. List all your collaborators if you have any.

I'm extremely busy this recently so I only assign 1 hour to this homework, and I think it might actually took less. Rate 2, 2.